

ENGINEERING REBOOT 2026  
Week 05 Master Workbook  
Version 1.0 - Week 5 Baseline

Document Status

Document Type: Master Workbook

Classification: Planning Authority

Version: 1.0

Status: Approved and Locked

Authority: Week 5 Planning Review

Workbook Purpose

The Week 05 Master Workbook serves as the planning and curriculum authority for Days 21–25. Daily Field Manuals shall be derived from this workbook. This workbook establishes weekly topics, exercises, projects, validation requirements, deliverables, and week completion criteria. Detailed engineering instruction, worked examples, implementation guidance, expected outputs, and engineering commentary are contained within the individual Daily Field Manuals.

Week Theme

Professional Python Foundations

Week Learning Outcomes

- Handle runtime exceptions gracefully.
- Validate user input before processing.
- Apply defensive programming techniques.
- Design reusable and maintainable functions.
- Use Python type hints to improve readability.
- Organize applications into professional multi-file projects.

Day 21 — Exception Handling Fundamentals

Concepts: Runtime Exceptions, try, except, else, finally, raise, Exception Hierarchy.

Exercises: `value_error_demo.py`, `zero_division_demo.py`, `multiple_exceptions_demo.py`, `try_else_demo.py`, `finally_demo.py`, `raise_exception_demo.py`, `nested_exception_demo.py`.

Project: Safe Calculator (`main.py`, `calculator.py`, `input_helpers.py`).

Validation: Handle invalid input, division by zero, multiple exceptions, recovery, expected outputs.

Deliverables: Exercise programs, Safe Calculator, Notes.md, Journal.md.

Day 22 — Input Validation & Defensive Programming

Concepts: Validation Loops, Range Checking, Type Validation, Defensive Programming, Input Sanitization.

Exercises: `numeric_validation.py`, `range_validation.py`, `menu_validation.py`.

Project: `validated_registration_system.py`.

Deliverables: Exercises, project, Notes.md, Journal.md.

Day 23 — Professional Function Design

Concepts: Single Responsibility Principle, Function Decomposition, Code Reuse, Documentation, Clean Interfaces.

Exercises: `function_refactor.py`, `helper_functions.py`, `utility_library.py`.

Project: `employee_payroll_system.py`.

Day 24 — Type Hints & Modern Python

Concepts: Type Annotations, Return Types, Optional, Union, Modern Python Style.

Exercises: `type_hints_demo.py`, `typed_functions.py`, `optional_examples.py`.

Project: `typed_inventory_manager.py`.

#### Day 25 — Program Organization

Concepts: Multi-module Applications, Package Organization, Configuration, Separation of Concerns.

Exercises: `config_demo.py`, `package_demo.py`, `module_structure.py`.

Project: Professional Console Application (`main.py`, `config.py`, `utilities.py`, `menus.py`).

#### Week Success Criteria

Complete Days 21–25, master exception handling, input validation, defensive programming, professional functions, type hints, and multi-module applications.

#### Week 06 Preview

Linux Fundamentals, WSL2, Bash Essentials, Linux Filesystem, File Permissions, Git under Linux, Python under Linux.

#### Approval Record

Review Status: Approved

Workbook Status: Locked

Effective Beginning: Day 21

Publication Status: Approved for Archival